

# Om Mishra

+91 7021552396 | omm512462@gmail.com | LinkedIn | GitHub

## EXPERIENCE

### FlyRank AI — Machine Learning Intern

06/2026 – Present Remote

- Worked on search-intelligence problems involving content performance, ranking, and content-refresh prioritization using real search and analytics data.
- Built and evaluated a Logistic Regression ranking model with client-level holdout validation, achieving **0.85 Precision@20** and **0.84 Precision@50** against a rule-based baseline.
- Applied data-contract, leakage-check, error-analysis, and permutation-importance techniques to validate model reliability and interpretability.

## PROJECTS

### Transformer Translation System



- Built and trained a complete Encoder-Decoder Transformer architecture from scratch for English-to-French neural machine translation on **2.3M sentence pairs** using PyTorch.
- Implemented SentencePiece tokenization, positional encoding, multi-head attention, masking, encoder-decoder cross-attention, beam search decoding, and autoregressive inference.
- Achieved a highly competitive **34.82 BLEU** score on **24k sentence validation set** and developed a Streamlit-based translation app with cross-attention heatmap visualizations.
- Improved training stability using Noam Learning Rate Scheduling, gradient clipping, custom validation pipelines, and numerical stability debugging during large-scale Transformer training.

### PulmoScan AI: High-Sensitivity Pneumonia Screening Tool



- Engineered a medical diagnostic system to detect pneumonia from chest X-rays, explicitly prioritizing **Recall (Sensitivity)** to minimize false negatives in clinical screening.
- Fine-tuned and benchmarked DenseNet121 against ResNet50, achieving **99.7% Recall** versus **97.0%**, reducing missed pneumonia cases from 11 to 1.
- Implemented Explainable AI (Grad-CAM) with bicubic interpolation to visualize lung opacities and deployed a Streamlit application with caching, reducing inference latency by approximately **80%**.

### Ramarttha: Semantic & Narrative Analysis of Shri Ramcharitmanas



- Built an end-to-end NLP system for semantic analysis of a low-resource classical text using Word2Vec & FastText, with a Unicode-native preprocessing pipeline and verse-level embeddings for semantic retrieval & narrative-aware analysis, deployed through a Streamlit app for real-time exploration of semantic relationships & concept similarity.

### Urban Mobility Optimization: Bike Demand Prediction



- Built an ML forecasting system for hourly bike demand, achieving **R2 = 0.935** & evaluated 7 regression models, selecting a tuned **LightGBM Regressor**, with cyclic feature engineering, log-transformation, & SHAP for model interpretability.

### Telecom Customer Churn Prediction



- Built a churn prediction pipeline on **7k+ customer records**, applying EDA, feature engineering, & comparative classification modeling with evaluating 8 ML models, selecting LR with **80.9% Recall & ROC-AUC = 0.75**, & deployed a Streamlit app for churn scoring, batch prediction, & retention-focused decision support.

## SUMMARY

Aspiring Machine Learning Engineer proficient in Python, Deep Learning, and Transformer architectures, focused on building end-to-end AI systems. Experienced in developing and deploying ML applications using PyTorch, Scikit-learn, and Hugging Face, with a strong emphasis on model evaluation, explainability, and real-world impact. Strong foundation in Data Structures, Algorithms, and software engineering best practices.

## EDUCATION

### Bachelor of Science in Information Technology

Smt. C.H.M. College

2024 – 2027 Ulhasnagar, Maharashtra

## SKILLS

### Machine Learning & Deep Learning

PyTorch, Scikit-learn, CNNs, Transfer Learning

### Data Analysis & Visualization

NumPy, Pandas, Matplotlib, Seaborn, Plotly

### Computer Vision & XAI

OpenCV, Grad-CAM, SHAP

### Deployment & Tools

Streamlit, FastAPI, Git, GitHub, HuggingFace Hub, VS Code

## CERTIFICATIONS

### Oracle AI Foundations Associate

Certified in Artificial Intelligence fundamentals, Machine Learning concepts, and responsible AI by Oracle.

### NASA Space Apps Challenge – Galactic Problem Solver

Recognized for contributing innovative solutions during NASA's global hackathon challenge.